**1** $y = (x+3)(x-4)$ **[M1]**

$x^2 - x - 12$ **[A1]**

*[Total Marks: 2]*

**2** $(x-4)(x+2)$ **[M1]**

$x^2 - 2x - 8$ **[A1]**

*[Total Marks: 2]*

**3** Uses $(x-p)(x-q)$ with $p = -4$ and $q = 3$: $(x+4)(x-3)$ **[M1]**

$x^2 + x - 12$ **[A1]**

*[Total Marks: 2]*

**4** Uses the intercepts to form $y = (x+4)(x-7)$ **[M1]**

$x^2 - 3x - 28$ **[A1]**

*[Total Marks: 2]*

**5** Uses the vertex form $y = (x-4)^2 - 3$ **[M1]**

Expands to $x^2 - 8x + 16 - 3$ **[M1]**

$b = -8$ and $c = 13$ **[A1]**

*[Total Marks: 3]*

**6** Uses the vertex form $y = (x-3)^2 - 4$ OR uses $-\dfrac{b}{2(1)} = 3$ **[M1]**

$b = -6$ **[A1]**

$c = 5$ **[A1]**

*[Total Marks: 3]*

**7** $y = (x-3)^2 - 4$ **[M1]**

$y = x^2 - 6x + 9 - 4$ **[M1]**

$x^2 - 6x + 5$ **[A1]**

*[Total Marks: 3]*

**8 (a)** $x^2 + bx + c = (x+4)(x-2)$ **[M1]**

$b = 2$, $c = -8$ **[A1]**

**(b)** Finds $x$-coordinate of vertex: $x = \dfrac{-4+2}{2} = -1$ **[M1]**

$(-1, -9)$ **[A1]**

*[Total Marks: 4]*

**9** **(a)** Writes $f(x) = (x+3)(x-5)$ **[M1]**

Expands correctly: $x^2 - 2x - 15$ **[M1]**

$b = -2$, $c = -15$ **[A1]**

**(b)** $(0, -15)$ **[B1]**

*[Total Marks: 4]*

**10** **(a)** Uses vertex form: $y = (x-3)^2 - 4$ **[M1]**

$x^2 - 6x + 5$ **[A1]**

**(b)** Sets $x^2 - 6x + 5 = 0 \implies (x-1)(x-5) = 0$ **[M1]**

$(1, 0)$ and $(5, 0)$ **[A1]**

*[Total Marks: 4]*

**11** **(a)** $h = 3$ **[B1]**

$k = -2$ **[B1]**

**(b)** $p = (5-3)^2 - 2$ **[M1]**

$p = 2$ **[A1]**

*[Total Marks: 4]*

**12** **(a)** $(x-3)^2$ seen **[B1]**

$(x-3)^2 - 4$ **[B1]**

**(b)** $y = x^2 - 6x + 9 - 4$ **[M1]**

$b = -6$ and $c = 5$ **[A1]**

**(c)** $(0, 5)$ **[B1]**

*[Total Marks: 5]*

**13** **(a)** Uses vertex form: $y = (x-2)^2 - 9$ or $x$-coordinate of vertex $\dfrac{-p}{2} = 2$ **[M1]**

Expands to find $p$ and $q$, or solves $2^2 + 2p + q = -9$ **[M1]**

$p = -4$ and $q = -5$ **[A1]**

**(b)** $x^2 - 4x - 5 = 0$ **[M1]**

$(x-5)(x+1) = 0$ **[M1]**

$A = (-1, 0)$ and $B = (5, 0)$ **[A1]**

*[Total Marks: 6]*

**14** $f(x) = a(x + 2)(x - 5)$ **[M1]**

$a(0 + 2)(0 - 5) = -20$ or $-10a = -20$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 6x - 20$ **[A1]**

*[Total Marks: 4]*

**15** $f(x) = a(x + 2)(x - 6)$ **[M1]**

$-24 = a(0 + 2)(0 - 6)$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 8x - 24$ **[A1]**

*[Total Marks: 4]*

**16 (a)** Writes $g(x) = a(x - 2)^2 - 9$ **[M1]**

Substitutes $(5, 9)$: $9 = a(5 - 2)^2 - 9$ or $9a = 18$ **[M1]**

$2(x - 2)^2 - 9$ **[A1]**

**(b)** $2(x^2 - 4x + 4) - 9$ **[M1]**

$2x^2 - 8x - 1$ **[A1]**

**(c)** $x = 2$ **[B1]**

*[Total Marks: 6]*

**17 (a)** Uses intercept form: $y = a(x + 1)(x - 4)$ **[M1]**

Substitutes $(0, -8)$ to find $a$: $-8 = a(0 + 1)(0 - 4) \implies -4a = -8 \implies a = 2$ **[M1]**

$2x^2 - 6x - 8$ **[A1]**

**(b)** Finds the $x$-coordinate of the vertex by symmetry: $x = \dfrac{-1 + 4}{2} = 1.5$ (or uses $x = -\dfrac{b}{2a}$) **[M1]**

Substitutes $x = 1.5$ into the equation: $y = 2(1.5)^2 - 6(1.5) - 8$ **[M1]**

$(1.5, -12.5)$ (or $(\dfrac{3}{2}, -\dfrac{25}{2})$) **[A1]**

*[Total Marks: 6]*

**18 (a)** $y = a(x + 2)(x - 4)$ **[M1]**

$16 = a(0 + 2)(0 - 4)$ **[M1]**

$a = -2$ **[A1]**

$y = -2x^2 + 4x + 16$ **[A1]**

**(b)** $x = \dfrac{-2+4}{2} = 1$ **[M1]**

$(1, 18)$ **[A1]**

**(c)** $x = 1$ **[B1]**

*[Total Marks: 7]*

**19 (a)** $h = -1$ **[B1]**

$k = 8$ **[B1]**

**(b)** $-10 = a(2-(-1))^2 + 8$ **[M1]**

$9a = -18$ **[M1]**

$-2$ **[A1]**

**(c)** $-2(x+1)^2 + 8$ **[M1]**

$-2(x^2 + 2x + 1) + 8$ **[M1]**

$-2x^2 - 4x + 6$ **[A1]**

*[Total Marks: 8]*

**20** $f(x) = a(x+1)(x-4)$ **[M1]**

$-8 = a(0+1)(0-4)$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 6x - 8$ **[A1]**

*[Total Marks: 4]*

**21 (a)** Forms the equation $f(x) = a(x+2)^2 + 8$ **[M1]**

Substitutes $(0, -4)$ into their equation:

$-4 = a(0+2)^2 + 8$ **[M1]**

$4a = -12 \implies a = -3$ **[A1]**

$-3x^2 - 12x - 4$ **[A1]**

**(b)** Sets $-3(x+2)^2 + 8 = 0$ or $-3x^2 - 12x - 4 = 0$ **[M1]**

Rearranges to $(x+2)^2 = \dfrac{8}{3}$ OR correctly substitutes into the quadratic formula:

$x = \dfrac{-(-12) \pm \sqrt{(-12)^2 - 4(-3)(-4)}}{2(-3)}$ **[M1]**

$\dfrac{-6 \pm 2\sqrt{6}}{3}$ **[A1]**

*[Total Marks: 7]*

**22** $y = a(x-3)^2 - 5$ **[M1]**

$7 = a(1-3)^2 - 5 \implies 12 = 4a \implies a = 3$ **[M1]**

$3(x-3)^2 - 5$ **[A1]**

*[Total Marks: 3]*

**23** $y = a(x+2)^2 + 5$ **[M1]**

$-13 = a(1+2)^2 + 5$ **[M1]**

$-18 = 9a$

a = -2$[M1]$

$-2(x+2)^2 + 5$ **[A1]**

*[Total Marks: 3]*

**24 (a)** Uses vertex form: $y = a(x-2)^2 + 7$ **[M1]**

Substitutes $(4, -5)$ into their equation: $-5 = a(4-2)^2 + 7$ **[M1]**

$-3(x-2)^2 + 7$ **[A1]**

**(b)** $-5$ **[B1]**

*[Total Marks: 4]*

**25** Writes the function in the factored form $y = a(x+1)(x-5)$ **[M1]**

Substitutes $(2, -18)$ into their equation:

$-18 = a(2+1)(2-5)$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 8x - 10$ **[A1]**

*[Total Marks: 4]*

**26** Uses vertex form $y = a(x-2)^2 - 5$ **[M1]**

Substitutes $(4, 7)$ into their equation: $7 = a(4-2)^2 - 5$ **[M1]**

$4a = 12$

a = 3$[A1]$

$3x^2 - 12x + 7$ **[A1]**

*[Total Marks: 4]*

**27 (a)** $h = 2$ **[B1]**

$k = 5$ **[B1]**

**(b)** Substitutes $(4, -7)$ into equation: $-7 = a(4-2)^2 + 5$ **[M1]**

$a = -3$ **[A1]**

**(c)** Substitutes $x = 0$ into $y = -3(x-2)^2 + 5$ **[M1]**

$(0, -7)$ **[A1]**

*[Total Marks: 6]*

**28** **(a)** Sets up the factored form $f(x) = a(x+2)(x-5)$ **[M1]**

Substitutes the point $(3, 10)$: $10 = a(3+2)(3-5)$ **[M1]**

Solves for $a$: $10 = a(5)(-2) \implies a = -1$ **[A1]**

$f(x) = -x^2 + 3x + 10$ **[A1]**

**(b)** Finds the $x$-coordinate of the vertex (e.g., midpoint of intercepts $\frac{-2+5}{2} = 1.5$ or $x = \frac{-b}{2a}$) **[M1]**

$(1.5, 12.25)$ **[A1]**

*[Total Marks: 6]*

**29** $y = a(x+1)^2 + 8$ **[M1]**

$-10 = a(2+1)^2 + 8$ **[M1]**

$9a = -18 \implies a = -2$ **[A1]**

$-2(x^2 + 2x + 1) + 8$ **[M1]**

$a = -2$, $b = -4$, $c = 6$ **[A1]**

*[Total Marks: 5]*

**30** **(a)** $y = a(x+3)(x-2)$ **[M1]**

$12 = a(0+3)(0-2)$ **[M1]**

$12 = -6a$

a = -2[$M1$]

$-2x^2 - 2x + 12$ **[A1]**

**(b)** $x = \frac{-3+2}{2}$ or $x = \frac{-(-2)}{2(-2)} = -0.5$ **[M1]**

$y = -2(-0.5)^2 - 2(-0.5) + 12 = 12.5$ **[M1]**

$(-0.5, 12.5)$ **[A1]**

*[Total Marks: 6]*

**31** **(a)** Sets up equation using $x$-intercepts: $f(x) = a(x+1.5)(x-4)$ OR $f(x) = a(2x+3)(x-4)$ **[M1]**

Substitutes $(2, -14)$ to find $a$: $-14 = a(2+1.5)(2-4)$ **[M1]**

Solves for $a$: $a = 2$ **[A1]**

$2x^2 - 5x - 12$ **[A1]**

**(b)** Identifies $x$-coordinate of vertex as midpoint of roots: $x = \dfrac{-1.5 + 4}{2} = 1.25$ OR uses $x = \dfrac{-b}{2a}$ **[M1]**

Substitutes $x = 1.25$ into their function: $y = 2(1.25)^2 - 5(1.25) - 12$ **[M1]**

$(1.25, -15.125)$ OR $\left(\dfrac{5}{4}, -\dfrac{121}{8}\right)$ **[A1]**

*[Total Marks: 7]*

**32** **(a)** **(i)** $h = -4$ **[B1]**

$k = 20$ **[B1]**

**(ii)** $0 = a(-2 + 4)^2 + 20$ **[M1]**

$a = -5$ **[A1]**

**(iii)** $-6$ **[B1]**

**(iv)** $f(x) = -5(x + 4)^2 + 20 = -5(x^2 + 8x + 16) + 20$ **[M1]**

$-5x^2 - 40x - 60$ **[A1]**

*[Total Marks: 7]*

**33** **(a)** Writes $h(x) = a(x + 2)(x - 4)$ **[M1]**

Substitutes $(1, 13.5)$: $13.5 = a(3)(-3) \implies -9a = 13.5$ **[M1]**

$a = -1.5$ (or $-\dfrac{3}{2}$) **[A1]**

Expands $-1.5(x^2 - 2x - 8)$ to obtain $b = 3$ and $c = 12$ **[A1]**

**(b)** $x$-coordinate of vertex $= \dfrac{-2 + 4}{2} = 1$ **[M1]**

$(1, 13.5)$ **[A1]**

**(c)** $h(x) \leq 13.5$ or $y \leq 13.5$ **[B1]**

*[Total Marks: 7]*

**34** **(a)** Sets up the equation in factored form: $f(x) = a(x + 1)(x - 5)$ **[M1]**

Substitutes $(0, 10)$ to find $a$: $10 = a(0 + 1)(0 - 5)$ **[M1]**

$a = -2$ **[A1]**

$-2x^2 + 8x + 10$ **[A1]**

**(b)** Finds the $x$-coordinate of the vertex using symmetry $x = \dfrac{-1 + 5}{2} = 2$ OR uses $x = -\dfrac{b}{2a}$ **[M1]**

$(2, 18)$ **[A1]**

**(c)** Translates the vertex $(2, 18)$

(4, 15)$OR applies transformation$-2((x-2)-2)$^2$ + 18 − 3 [M1]

$-2(x-4)^2+15$ **[A1]**

*[Total Marks: 8]*

**35** **(a)** Uses the vertex coordinates to write $f(x)=a(x+2)^2+8$ **[M1]**

Substitutes the $y$-intercept $(0,5)$: $5=a(0+2)^2+8$ **[M1]**

$f(x)=-0.75(x+2)^2+8$ or equivalent using fractions like $-\frac{3}{4}$ **[A1]**

**(b)** Expands $(x+2)^2$ as $x^2+4x+4$ **[M1]**

$f(x)=-0.75x^2-3x+5$ **[A1]**

**(c)** Equates function to zero: $-0.75(x+2)^2+8=0$ or $-0.75x^2-3x+5=0$ **[M1]**

Rearranges to $(x+2)^2=\frac{32}{3}$ or correctly applies the quadratic formula **[M1]**

$x=1.27$ and $x=-5.27$ **[A1]**

*[Total Marks: 8]*

**36** **(a)** $c=15$ **[B1]**

$y=a(x+2)(x-5)$ **[M1]**

$15=a(0+2)(0-5)$ **[M1]**

$a=-1.5$ **[A1]**

$b=4.5$ **[A1]**

**(b)** $x=\frac{-2+5}{2}=1.5$ **[M1]**

$y=-1.5(1.5)^2+4.5(1.5)+15$ **[M1]**

$(1.5, 18.375)$ or $\left(\frac{3}{2}, \frac{147}{8}\right)$ **[A1]**

**(c)** $k>18.375$ **[B1]**

*[Total Marks: 9]*

**37** **(a)** Uses vertex form: $y=a(x+2)^2+18$ **[M1]**

Substitutes $(0,10)$ to find $a$: $10=a(0+2)^2+18 \implies 4a=-8 \implies a=-2$ **[M1]**

$-2x^2-8x+10$ **[A1]**

**(b)** Sets $f(x)=0$: $-2x^2-8x+10=0$ **[M1]**

$x^2+4x-5=0 \implies (x+5)(x-1)=0 \implies x=-5$ or $x=1$ **[M1]**

6 **[A1]**

**(c)** Equates line and curve: $-2x^2-8x+10=-2x+k$ **[M1]**

Rearranges to form a quadratic equation: $2x^2+6x+(k-10)=0$ (or equivalent) **[M1]**

Sets discriminant $\Delta = 0$ for a tangent: $6^2 - 4(2)(k-10) = 0$ **[M1]**

$14.5$ (or $\frac{29}{2}$) **[A1]**

*[Total Marks: 10]*

**38** **(a)** $h = 4$ and $k = -5$ **[B1]**

$8.5 = a(1-4)^2 - 5$ **[M1]**

$a = 1.5$ **[A1]**

**(b)** $1.5(x-4)^2 - 5 = 0$ or $(x-4)^2 = \frac{10}{3}$ **[M1]**

$x = 4 \pm \sqrt{\frac{10}{3}}$ **[M1]**

$x = 2.17$ and $x = 5.83$ **[A1]**

**(c)** $1.5(x-4)^2 - 5 = 2x - 7$ **[M1]**

$3x^2 - 28x + 52 = 0$ **[A1]**

$x = 2.56$ and $x = 6.77$ **[A1]**

**(d)** New vertex at $(1, 4)$ **[M1]**

$1.5(x-1)^2 + 4$ **[A1]**

*[Total Marks: 11]*

**39** **(a)** Forms equation using $x$-intercepts: $f(x) = a(x+1)(x-3)$ OR using vertex $f(x) = a(x-1)^2 - 8$ **[M1]**

Substitutes appropriate point to find $a$: $a(1+1)(1-3) = -8 \implies -4a = -8 \implies a = 2$ **[M1]**

Expands to show final form: $2(x^2 - 2x - 3) = 2x^2 - 4x - 6$ **[A1]**

**(b)** $2x^2 - 4x - 6 = 10 \implies 2x^2 - 4x - 16 = 0$ **[M1]**

Factors correctly to $(x-4)(x+2) = 0$ OR uses quadratic formula correctly **[M1]**

$x = 4$ and $x = -2$ **[A1]**

**(c)** Sets equations equal: $2x^2 - 4x - 6 = mx - 8$ leading to $2x^2 - (4+m)x + 2 = 0$ **[M1]**

States condition for tangent (discriminant $= 0$): $(-(4+m))^2 - 4(2)(2) = 0$ **[M1]**

Simplifies to $(4+m)^2 = 16$ or $m^2 + 8m = 0$ **[M1]**

$m = 0$, $m = -8$ **[A1]**

**(d)** Recognises minimum $y$-value must be strictly greater than 0: $-8 + k > 0$ **[M1]**

$k > 8$ **[A1]**

*[Total Marks: 12]*